19.3 An electron passing through an apparatus is un-deflected. Does that imply that the apparatus has no magnetic field? Explain.

19.4 A particle with charge 10^{-6}C is moving in +y direction at 10^4 m/s at right angles to a magnetic field of 2 T (Fig. 19.33). (a) Find the magnitude and the direction of the force on the particle. (b) What would the force be if the velocity were in the –y direction?

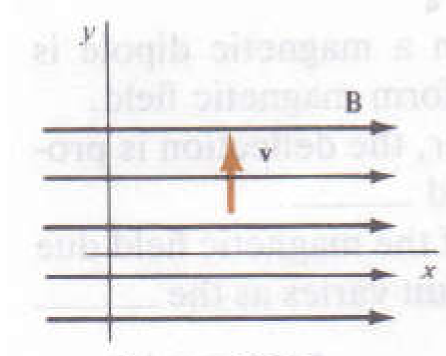

Figure 19.33.

19.7 An object of mass 0.01kg moves at 100m/s at angle of 30^o to a magnetic field of 10^{-2}T (Fig. 34). If its charge is -10^{-3}C, find the magnitude and the direction of (a) the magnetic force (b) the acceleration.

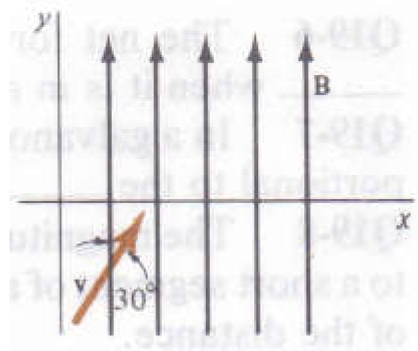

Figure 19.34.

19.12 A power line carries a current of 1000A. The earth's magnetic field makes an angle 73^o with the line has a magnitude of 7×10^{-5}T. Find the magnitude of the magnetic force on a 100m long section of the line.

19.13 (a) Find the magnitude and direction of the force on each of the three straight segments in Fig. 19.35. (b) What is the net force on the circuit?

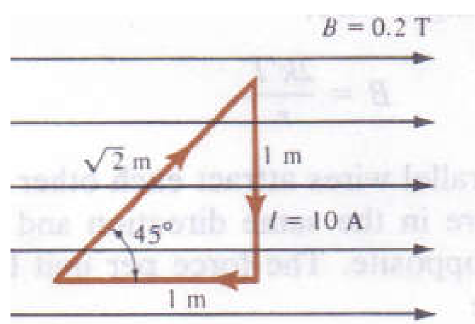

Figure 19.35.

19.21 Fifty circular turns of wire are wound closely together, so that all of them have a radius of close to 0.05m. If the wire carries a current of 2A, what is the magnitude of the magnetic field at its center?

19.25 Along, straight wire carries a current of 4A. Find the magnitude and direction of the field at a distance of (a) 0.1m (b) 1m

19.30 Three parallel wires lie in a plane. The separation between adjacent wires is 0.1m, and each wire carries a 10A current in the same direction. Find the magnitude of the net force per unit length on (a) the central wire (b) one of the outer wires.

19.43 According to Bohr model, in the normal hydrogent atom the electron orbits the proton in a circle of radius 5.1×0^{-11}m at a frequency of 6.8×10^{15}Hz. (a) What current due to the orbital motion of the electron? (b) What is the magnetic field at the proton due this current?

19.47 A beam of particles with charge +e moves in a circle of radius 3m in a magntic field of magnitude 0.2T. If the particles are protons (a) what is their velocity? (b) what is the kinetic energy of the particle?

Q(1) Three particles are moving perpendicular to a uniform magnetic field and travel on a circular paths as shown in Figure. The particles have same mass and speed. List the particles in order of their charge magnitude, largest to smallest.